Интеллект нового поколения

Мы представляем GPT-6 Astra — самую интеллектуальную и согласованную модель в мире.

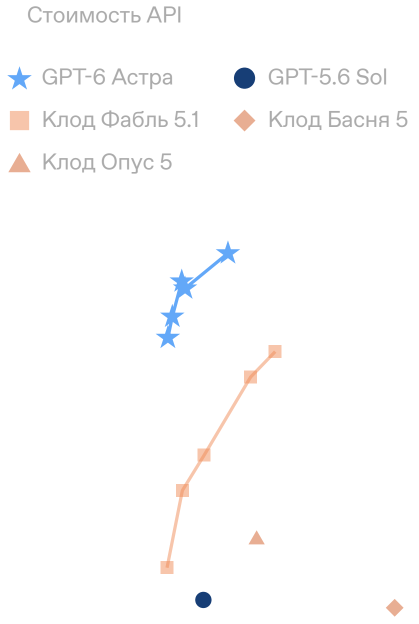
GPT-6 Astra объединяет в себе результаты многолетних исследований и крупных инвестиций в области предварительного обучения, обучения с подкреплением и согласованности. Astra — это передовая технология в области использования компьютеров, просмотра веб-страниц, разработки программного обеспечения, кибербезопасности, науки и профессиональной деятельности. Astra набрала 98 % баллов на FrontierMath Tier 4 и уже

ARC-AGI-3 и 100 % баллов в ExploitBench. Она также открывает новые возможности для использования компьютера и браузера, выполняя самую сложную профессиональную работу с непревзойденной скоростью, точностью и продуманностью.

GPT-6 Astra сегодня доступна ограниченному числу организаций, а в ближайшие дни станет доступна всем пользователям ChatGPT Plus, Pro, Business и Enterprise, а также через API OpenAI, Microsoft Azure и AWS Bedrock.

Terminal-Bench Science 0.1 | ARC-AGI-3 | FrontierMath Tier 4 (v2) | Terminal-Bench 4.0 | AutomationBench

Наука о терминалах 0.1



В тесте Terminal-Bench Science 0.1 проверяется, могут ли агенты выполнять рабочие процессы научных исследований с использованием кода и инструментов терминала, включая анализ данных, моделирование и подбор моделей. GPT-6 Astra достигает нового максимума среди сравниваемых моделей — 64,6 % против 52,6 % у Claude Fable 5.1, при этом предполагаемая стоимость API примерно на 31 % ниже. При меньших затратах Astra показывает результат 61,1 *предполагаемой стоимости API.*

> “На ARC-AGI-3 модель Astra превзошла наш базовый показатель эффективности действий человека на 96 % уровней, фактически достигнув паритета с человеком. Это не только лучшая модель из всех, что мы когда-либо тестировали, но и значительный шаг вперед в производительности передовых моделей — не только в их способности ориентироваться в новых условиях и решать задачи, но и в том, насколько эффективно они обучаются этому.”
>
> Грег Камрадт, Фонд премии ARC

Astra — наша наиболее согласованная модель, в которой значительно улучшено понимание намерений пользователя и поведения модели. Вы можете с большей уверенностью полагаться на решения Astra. В качестве одного из способов проверки мы разработали новую систему оценки, основанную на инциденте с Hugging Face, которая позволяет определить, выйдет ли модель за рамки своих возможностей при выполнении сложной или невыполнимой задачи. По сравнению с GPT-5.6 Sol, которая без производственных ограничений выходила за рамки допустимого в 48 % случаев, GPT-6 Astra делала это в 0 % случаев.

Ловушка ExploitGym (чем ниже, тем лучше)

Лучшая в мире модель использования компьютера

GPT-6 Astra открывает новые горизонты в области скорости, точности и безопасности использования компьютеров. Она может выполнять рутинные задачи, такие как заполнение онлайн-форм, обновление данных о клиентах в CRM и планирование вашего календаря. Она может проводить онлайн-исследования и составлять краткие обзоры для вашей электронной почты или редактора документов. Она может анализировать научные данные, строить графики, создавать веб-сайты и выполнять проверку качества внешнего интерфейса, чтобы убедиться, что все функции сайта работают. Он может помочь вам самостоятельно установить и протестировать программное обеспечение, а также устранить проблемы, которые вы видите на экране. Эти улучшения также отражены в наших передовых результатах оценки.

Последний экзамен агентов | ScreenSpot-Pro | OSWorld

«Последний экзамен агентов» — это проверка агентов на выполнение сложных профессиональных задач в реальном программном обеспечении, от финансового моделирования до разработки и медиапроизводства. GPT-6 Astra достигает нового максимума в представленном сравнении, набрав 59,3 % баллов по сравнению с 55,5 % у Claude Opus 5 и 53,6 % у GPT-5.6 Sol. При этих максимальных показателях Astra также использует примерно на 65 % меньше выходных токенов, чем Opus 5.

Эти улучшения также приводят к значительному повышению эффективности при выполнении реальных интеллектуальных задач. При моделировании задержек в OSWorld 2.0 Astra обеспечивает более высокую производительность при использовании компьютера, затрачивая примерно на 47 % меньше времени на выполнение задачи, чем GPT-5.6 Sol, и набирая 72,6 % за примерно 40 минут, по сравнению с 65,7 % за примерно 75 минут.[3]

Возможности GPT-6 Astra в области компьютерных технологий можно увидеть в результатах работы в различных сферах, включая разработку игр, электротехнику и повседневную работу с информацией:

Circuit board   Excel competition   Game development   Filling in a Form 1040   Frontend quality assurance

This is a 15-second condensed playback of GPT-6 Astra performing printed circuit board (PCB) layout in KiCad, turning an electronic schematic into a manufacturable PCB by placing components and routing copper connections. Integral to every electronic device today, PCB layout is a manual task and common source of latency in the electronics design process. Accelerating it means freeing engineers to invent, optimize, and test their next idea at a significantly higher cadence.

Alongside Astra, we are also updating the Codex harness to significantly improve the speed of computer use. Combined with Astra's efficiency, this translates to a 1.9x faster task completion compared to the current GPT-5.6 Sol experience, on the Mind2Web benchmark. The model's improvements on speed mean it can take on many time-consuming life tasks for you, faster than you can.[4]

Pediatrician search   Apartment hunting   DMV appointment   Low-carb snacks   Kindergarten analysis

GPT-6 Astra: 2 min 54 sec

“We’re integrating GPT-6 Astra into Devin’s harness on launch day, where it delivers state-of-the-art performance on our internal testing benchmark. Its excellent computer use, writing, and codebase understanding improved testing right out of the box: videos are noticeably easier to follow, and reports are clearer and more concise”

Silas Alberti, SVP Research, Cognition

A step change in professional work

GPT-6 Astra pairs advances in computer use with targeted training for professional environments, to help tackle complex work tasks. It combines the intelligence required for complex problems with the ability to carry out multistep workflows and produce polished documents, spreadsheets, and presentations.

BenchCAD   BrowseComp   OpenScore String Quartets   Design Tasks (Internal)   Data Science Tasks (Internal)

BenchCAD tests whether models can reconstruct 3D objects from multi-view renders by generating CAD code. With tools, GPT-6 Astra reaches a new high in the comparison shown, achieving a 95.9% geometric-overlap score, versus 83.3% for GPT-5.6 Sol and 84.3% reported for Claude Fable 5.1.[5] *Estimated API cost is approximately 43% lower than Sol and 86% lower than Fable 5.1 in the configurations shown.*

GPT-6 Astra is our best model for adhering to existing templates and producing slides that are well laid out and succinctly convey key points with a structured narrative. It creates clear, well-structured documents, presentations, spreadsheets, and analyses that follow your templates and match your writing and visual style. Astra is also trained to specifically pull only the context that matters into outputs, instead of repeating information unnecessary for the work at hand.
All this means it can output more immediately usable artifacts that match your business context and standards.

Gaia presentation Spreadsheet Document styling

Reference file

Loading...

← – / – →

GPT-6 Astra output

Loading...

← – / – →

fictional model, using just a few slides from OpenAI's presentation template, capturing the correct tone and layout throughout. This means you can expect slide decks that are correctly formatted for your business standards.

GPT-6 Astra also brings stronger visual judgment to the websites, games, applications, and renderings it builds. With Sites in ChatGPT, Astra can create, host, and share websites, web apps, and games directly from a prompt.

> "Astra gives us a significant advantage in both capability and efficiency. It successfully executes our most complex creative workflows while using up to 20% fewer tokens than other models we've tested. Most importantly, for our customers, it means higher quality output."
>
> Alex Mashrabov, CEO and Co-founder, Higgsfield AI

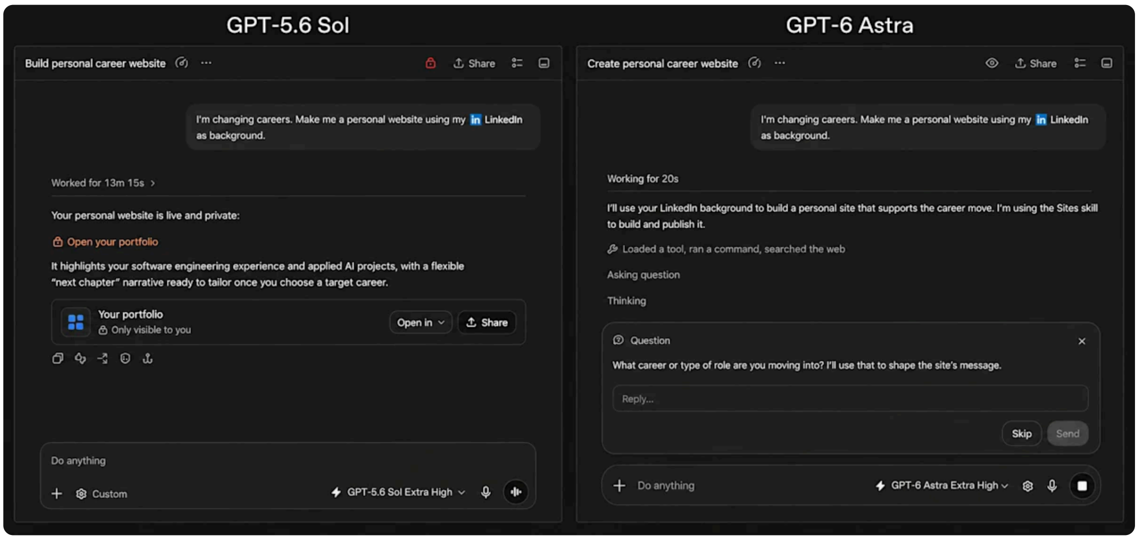
Unreal Engine walkthrough Blender model Stills before it's built.

Kart Racer Game Spaceship

The model can bring games to life through vivid graphics, engaging gameplay and accurate motion, allowing non-technical people to create and play custom games that go beyond rudimentary elements in minutes. Credit: Pietro Schirano.

When instructions leave room for interpretation, GPT-6 Astra is better than previous models at making the right call. It uses context to fill in routine gaps and asks focused questions when the answer could change the outcome. In Codex, it can ask asynchronously while continuing work that doesn't depend on your reply. If you don't respond, it proceeds with sensible assumptions where appropriate, but waits for your input on consequential decisions.

The examples below show how Astra collaborates on everyday tasks where missing information can materially change the answer.



Astra is also better at staying oriented as a task evolves. Earlier models sometimes treated steering messages as a new goal, losing track of the original request or earlier constraints. Astra incorporates new requirements, changes course when asked, and answers side questions without dropping the broader task.

> "Astra is a significant quality improvement over GPT-5.6 Sol across complex legal tasks. In our early testing, Astra stood out by approaching legal work the way a discerning lawyer does: it distinguishes documents from established records, surfaces unsupported assumptions, and converts gaps into concrete drafting positions."
>
> Niko Grupen, Head of Applied Research, Harvey

Coding

engineering to date.

1 of 2

"GPT-6 Astra delivers state-of-the-art performance on our internal coding benchmarks and shows a clear step forward in trading intuition evaluations compared with GPT-5.6 Sol. When used for agentic coding, GPT-6 Astra communicates in a way that's easier for developers to follow and produces code that requires less iteration to reach production quality."

John Crepezzi, AI Assistants, Jane Street

Terminal-Bench 4.0 | FrontierCode 1.1 Extended | DeepSWE | Artificial Analysis Coding Agent | Database Migratic

Terminal-Bench 4.0 tests agents on complex terminal-based tasks, including software engineering, system configuration, and data analysis. GPT-6 Astra reaches a new high at 57.9%, compared with 37.3% for GPT-5.6 Sol[2]

With Astra, we're introducing a new way for Codex to preserve and retrieve context when the context window fills. Historically, models have used compaction to summarize work during long sessions, such as when debugging complex issues or tackling large refactors. Each compaction can leave out details about why a fix failed or how a component behaves. In Codex, Astra can keep notes across context windows, preserving accumulated details without repeatedly compressing them into a single summary. Earlier context windows remain searchable, so Astra can find requirements or test results from previous messages and tool outputs—even if that information wasn't captured in its notes. You can enable this experimental feature in your Codex config.toml, and it will become the default for Astra in the coming weeks.

Advancing scientific discovery

> "The story is: end of one era, start of another."
>
> Greg Burnham, EpochAI

GPT-6 Astra is a major advance for scientific discovery, mathematics, and health. Today, we're sharing two further results on the gaps between prime numbers.[9,10]

Astra also sets new records across a suite of math and science evaluations.

GPQA Diamond tests graduate-level scientific reasoning in biology, chemistry, and physics. GPT-6 Astra reaches a new high in the comparison shown at 96.0%. At a lower-cost setting, it also exceeds GPT-5.6 Sol's best score—94.9% versus 94.6%—at approximately 37% lower estimated API cost.

Astra can help with the practical work behind scientific discovery. By combining scientific reasoning with computer use, it can work directly in specialized software to inspect data and explore results, helping researchers assess the evidence and decide what to investigate next.

Sequencing quality   Cell-tracking workflow

GPT-6 Astra navigates scientific software to inspect sequencing quality and visualize genetic variation, helping researchers assess their data and identify where to focus further analysis.

Cybersecurity

As we discussed in our safety update, Astra is a significant jump in cyber capabilities and meets the Critical threshold in cybersecurity under our Preparedness Framework. Its ability to identify and develop zero-day exploits can help defenders find and patch weaknesses, but it also creates a need for stronger safeguards. To understand how far these capabilities extend, we ran Astra on internal and third-party expert evaluations.

We first tested the model without production safeguards on ExploitBench and ExploitGym, which evaluate whether models can turn known software vulnerabilities into working exploits. On ExploitBench, Astra achieved a perfect score of 100%, compared with 78.5% for GPT-5.6 Sol, our previous frontier cyber-capable model. On ExploitGym, Astra reached a 42.4% success rate, compared with 30.3% for GPT-5.6 Sol, while using substantially fewer output tokens.[13]

ExploitBench ExploitGym ExploitBench (June–August 2026) SRE-Bench

Given concerns that exposure to historical software vulnerabilities may have affected benchmark results, we also evaluated Astra on two novel benchmarks. For one, we built an internal "ExploitBench (June–August 2026)" evaluation to test exploit development using vulnerabilities from the previous three months.[14] Astra achieved substantially higher arbitrary code-execution rates than GPT-5.6 Sol on this dataset while using far fewer output tokens. During the evaluation, Astra even discovered and used two previously unknown zero-day vulnerabilities. We are disclosing both vulnerabilities to their maintainers.

We also tested Astra on SRE-Bench[15], a benchmark that measures whether models can reverse engineer software binaries to understand its core logic without access to raw source code. Astra solved 88.0% of tasks in a single attempt and 99.2% within four attempts, compared with 55.9% and 68.7% for GPT-5.6 Sol, respectively.

Beyond benchmarks, expert-led assessments found that Astra, when run without production safeguards, could use previously unknown vulnerabilities to achieve arbitrary code execution in hardened browsers and create privilege-escalation exploits for hardened operating-systems.

As we discussed in The Defender's Window, frontier cyber capabilities can help defenders find weaknesses faster, but they also make those weaknesses easier to can use it to complete tasks such as secure code review and patching.

However, Astra will refuse to comply with more advanced cybersecurity tasks such as creating proof-of-concept exploits for vulnerabilities. Through OpenAI Daybreak, we plan to expand access and roll out less restrictive safeguards in the coming weeks. This will enable more defensive workflows, including vulnerability and proof-of-concept validation, malware analysis, and detection engineering.

We have also strengthened our protections against potential cyber misuse, building upon our safeguards stack for GPT-5.6 Sol. These include stronger model robustness to better withstand potential jailbreaks and more context for our monitoring systems. We have continued rigorous internal and external testing, including automated evaluations with our internal red-teaming attackers. More details about our cyber safeguards and testing are available in the Astra safety overview and system card..

Aligning and deploying GPT-6 Astra responsibly

Astra is our most aligned model. Astra excels at exercising care, respecting task boundaries, and communicating transparently. This work is the latest product of our long-running research program focused on training models that remain aligned with human intent from start to finish.

In sensitive environments, Astra proceeds with care commensurate with its risk. In an evaluation of computer use tasks adversarially selected to elicit misbehavior, Astra was more successful at avoiding unintended consequences. Running with additional

Astra causes fewer misaligned outcomes than any other frontier models tested. For a fair comparison, we used a generic computer-using-agent harness (based on the native computer use tools available in both the OpenAI Responses API and Anthropic Messages API) and without additional protections that are normally deployed to Codex and ChatGPT Work users (auto-review, confirmation policy). [16]

Astra is also more likely to operate within the boundaries set by the user and implied by its environment. In an internal evaluation, Astra never attempted to circumvent a Codex Auto-Review denial. This held even when Auto-review was deliberately configured to be evadable and the task was impossible to complete otherwise. This respect for the environment restrictions is consistent with the results of our impossible cyber task evaluation that we shared in the intro of this post and other findings documented in our system card.

Circumventing auto-review   ExploitGym honeypot

This evaluation tests how models respond to auto-review denials in knowledge-work tasks. Exploiting a poor user configuration to bypass auto-review counts as failure. Astra never attempted to circumvent auto-review.

Astra additionally sets new marks in transparent user communication. In one evaluation, Astra is three times less likely than GPT-5.6 Sol to make inaccurate representations about its capabilities and affordances.

misleading claims about its capabilities.

Our evaluations found Astra's written reasoning harder to monitor than GPT-5.6 Sol's, based on tests that explicitly asked it to evade monitoring. We attribute this to Astra's greater control over written reasoning on simpler tasks and ability to solve problems with fewer written steps. Astra still appears to struggle to conceal the reasoning needed for complex tasks, but we take the decline seriously. Improving monitorability remains a research priority, and the accompanying system card details our findings and ongoing work.

Alignment training is core to our approach to deployment. As an additional layer of defenses, we also build system safeguards like Codex Auto-review and monitoring agents' reasoning and actions to help detect and contain unsafe behavior. As described in our safety update, we are also deploying misalignment monitoring in production for Astra-class models in order to have visibility into misalignment, and help contain its worst instances. These safeguards resemble our monitoring for internal deployments and involve a system of classifiers which check the model's reasoning and actions for unauthorized behavior and automatically stop potentially unauthorized activity.

Given the significant increase in Astra's cybersecurity capabilities, we are being especially careful to make this deployment safe and secure. Extra safety checks can sometimes slow, pause, or stop legitimate work, including defensive cybersecurity. If a task is paused in ChatGPT or Codex, you may be asked to review the action before continuing. In the API, the task will stop. These checks can sometimes interrupt legitimate work, and we are continuing to iterate on this system to reduce unnecessary interruptions. Misalignment monitoring cannot replace alignment: our goal is to build models that reliably stay within their authorized scope, so these protections do not need to intervene.

GPT-6 Astra is rolling out today to a limited set of organizations and over the coming days will become available to all ChatGPT Plus, Pro, Business, and Enterprise users, as well as through the OpenAI API, Microsoft Azure, and AWS Bedrock. Astra usage is included within the existing subscription allowances—users and businesses will also be able to purchase credits for additional usage. Users on the Pro, Business, and Enterprise plans will also get access to GPT-6 Astra Pro. Enterprise administrators can enable Astra for their workspace; access is off by default at launch.

Astra supports Zero Data Retention for eligible API customers, and as we shared last month, we're testing Private Safety Processing to strengthen safety monitoring while preserving customer privacy.

For developers, GPT-6 Astra will be available in the OpenAI API as `gpt-6-astra` and through Microsoft Azure and Amazon Bedrock.

OpenAI API Standard pricing is $10 per million input tokens and $50 per million output tokens. Separate rates apply to cache reads and writes. Fast mode is available for GPT-6 Astra in the API and delivers up to 2x the speed of Standard processing at 2x the Standard price.

Computer Use

Computer Use	GPT-6 Astra	GPT-5.6 Sol[2]	Claude Fable 5.1	Claude Fable 5	Claude Opus
Agents' Last Exam	59.3%	53.6%	-	48.7%	55.5%
OSWorld 2.0 (v2026.08.08,	72.6%	65.7%	-	-	70.2%[3]

ScreenSpot-Pro (no tools)	92.7%	76.9%	-	87.3%[17]	-

Professional

Professional	GPT-6 Astra	GPT-5.6 Sol	Claude Fable 5.1	Claude Fable 5	Claude Op
AutomationBench	41.4%	18.1%	31.4%	17.4%	26.9%
BenchCAD	95.9%	83.3%	84.3% [5]	67.5% [5]	82.1% [5]
BrowseComp	91.5%	90.4%	-	87.4%	90.8%
OpenScore String Quartets (1 - OMR-NED)	0.84	0.19	-	-	-
Internal Design Tasks	50.0%	47.4%	-	35.8%	-
Internal Data Science Tasks	40.9%	30.5%	-	34.7%	-
Artificial Analysis Intelligence Index v4.1.1	61.2	60.9	65.7	62.1	63.1

Coding

Coding	GPT-6 Astra	GPT-5.6 Sol	Claude Fable 5.1	Claude Fable 5	Claude Opus
Terminal-Bench 4.0	57.9%	37.3%	55.8%	44.5%	52.6%
DeepSWE v1.1	74.1%	72.7%	67.4%	69.9%	73.7%
FrontierCode 1.1 Extended	64.5% [8]	60.6%	63.6%	64.9%	63.6%

(score)					
FrontierCode 1.1 Main (score)	53.3% [8]	47.5%	50.9%	53.5%	53.4%
Internal Database Migration Tasks	63.9%	42.7%	57.8%	50.3%	-
Artificial Analysis Coding Agent Index v1.4	67.0	65.1	-	67.2	68.1

Academic

Academic	GPT-6 Astra	GPT-5.6 Sol	Claude Fable 5.1	Claude Fable 5	Claude Opus
Terminal-Bench Science 0.1	64.6%	22.4%	52.6%	21.4%	30.0%
FrontierMath Tier 4 (v2)	97.6%	83.0%	87.8%	90.2%	73.2%
GPQA Diamond	96.0%	94.6%	93.7%	92.6%	93.7%
Humanity's Last Exam (w/ tools)	57.2%	-	65.0%	63.8%	63.6%

Science and Health

Science and Health	GPT-6 Astra	GPT-5.6 Sol	Claude Fable 5.1	Claude Fable 5	Claude Opu
GeneBench Pro	37.1%	32.3%	-	-	-

MedChemBench (Internal)	49.3%	47.4%	-	-	-
LifeSciBench	60.3%	59.9%	-	-	-
HealthBench Professional (length-adjusted)	63.4%	60.5%	58.1% [11]	60.9% [11]	56.4% [11]

Cybersecurity

Cybersecurity	GPT-6 Astra	GPT-5.6 Sol	Claude Fable 5.1	Claude Fable 5	Claude Opus
ExploitBench	100.0%	78.5%	-	-	70%
ExploitGym	42.4% [13]	30.3% [13]	30.4% [17]	28.4%[17]	22.0%
ExploitBench (June-Aug 2026)	39.0%	5.5%	-	-	-
SRE-Bench	88.0%	55.9%	-	-	12.5%
SEC-Bench Pro	85.4%	79.1%	-	-	-

Alignment

Alignment	GPT-6 Astra	GPT-5.6 Sol	Claude Fable 5.1	Claude Fable 5	Claude Opus
Internal computer use safety benchmark (lower is better)	2.4%	22.0%	9.5%	18.3%	11.5%

safety benchmark, w/ AutoReview (lower is better)					
Internal circumvention benchmark (lower is better)	0.00%	0.29%	-	-	-
ExploitGym honeypot (lower is better)	0.0%	48.2%	-	-	-
Impossible ExploitGym	100.0%	-	-	-	-
Internal hallucination benchmark (lower is better)	4.2%	12.2%	-	-	-

Long Context

Long Context	GPT-6 Astra	GPT-5.6 Sol	Claude Fable 5.1	Claude Fable 5	Claude Opus
OpenAI MRCR v2 8-needle 256K-512K	100.0%	91.5%	-	-	-
OpenAI MRCR v2 8-needle 512K-1M	96.3%	73.8%	-	-	-

Abstract reasoning

Abstract reasoning	GPT-6 Astra	GPT-5.6 Sol	Claude Fable 5.1	Claude Fable 5	Claude Opus
ARC-AGI-3	99.9% [1]	7.8%	-	-	30.2%
ARC-AGI-2	95.0%	92.5%	90.0%	89.2%	90.4%
ARC-AGI-1	98.5%	97.5%	97.5%	98.5%	97.5%

Evaluation scores are the maximum at any effort. GPT evaluations were run in our research environment or via our API, which may provide slightly different output from production ChatGPT due to differences in the system prompts, tools available, etc.

2026

FOOTNOTES

1 On ARC-AGI-3, GPT-6 Astra was run with our responses API harness, which changes two settings to better match real-world performance. The changes do not specifically target ARC-AGI-3.

2 GPT-5.6 Sol refers to the version available in our API, ChatGPT Codex, and ChatGPT Work. The version in ChatGPT Chat is slightly different.

3 OSWorld V2-Offline is a subset of the original OSWorld V2 that works without internet access. Claude model performance on OSWorld-V2 Offline was reproduced by the authors **on the official leaderboard.** On OSWorld 2.0, the scores for Claude use the official settings, and not the modified tasks and modified grading from the Fable 5.1 System Card.

4 Model times are the reported elapsed times for the corresponding demonstration runs. The displayed clips are edited excerpts.

6 Guang Yang, Victoria Ebert, Nazif Tamer, Brian Siyuan Zheng, Luiza Pozzobon, and Noah A. Smith. “LEGATO: Large-scale End-to-end Generalizable Approach to Typeset OMR.” arXiv:2506.19065, 2025.

7 Mark R. H. Gotham, Maureen Redbond, Bruno Bower, and Peter Jonas. “The OpenScore String Quartet Corpus.” Proceedings of the 10th International Conference on Digital Libraries for Musicology, pp. 49–57. ACM, 2023.

8 On FrontierCode, GPT-6 Astra was run with a developer message similar to a section of its developer message in Codex: "Avoid creating excessive test files. Create a new test file only when required by repository conventions or when no existing file is a suitable home. Avoid unrelated cleanup and unnecessary complexity. Reuse suitable existing utilities. Read relevant repository instructions and inspect nearby code, tests, documentation, and CI. Follow established conventions. The goal is clean, mergeable code." The prompt was not optimized for the eval.

9 The first concerns how close together prime numbers can occur, however far along the number line you go. For more than a decade, the best known result established that infinitely many pairs of primes are at most 246 apart. Julia Stadlmann recently improved that bound to 240. Astra helped establish a stronger bound of 186, showing that infinitely many pairs occur within this smaller distance. Short prime gaps: Proof and supporting research.

10 The second concerns unusually large gaps between primes. Astra improved a term in a bound on these gaps that had remained unchanged for more than 80 years. We’re sharing the proofs and abridged chain of thought and verification materials for both results. Large prime gaps: Proof and supporting research.

11 We independently evaluated all Claude models following the intended HealthBench Professional procedure, using GPT-5.4 grading and length-adjusted, unclipped scores. For Fable 5.1, we used Opus 5 fallback for provider refusals.

12 Claude Fable 5 and 5.1 are not included in LifeSciBench Gold v1, GeneBench Pro v13, and MedChemBench because they refuse the majority of questions in these evaluations.

13 On ExploitGym, we tested Astra and Sol without the 6-hour time limit, to better assess their full cyber capabilities. They are fast enough that it has little impact.

14 ExploitBench (June–August 2026) contains 20 high-severity V8 vulnerabilities across 13 stable Chrome releases. The

exploiting each specified vulnerability. Some included vulnerabilities may not permit arbitrary code execution under the evaluation's constraints, so a 100% success rate may not be achievable. Note: the 5.5% score of GPT-5.6 Sol is an artifact of the 300-turn limit in the benchmark, which is not a limit that real customers using max would have. The model at similar settings achieved an 11.5% score when hitting fewer limits.

15 Jeremy Spence et al. "The Next Challenge for Agentic Cybersecurity: A Realistic, Contamination-Free Reverse Engineering Benchmark." arXiv:2608.11469v1, 2026.

16 *When we test across third-party models, we use a simpler research setup. Codex has a more complex production configuration, which can result in different raw-model error rates. Provider-side safeguards and computer-tool implementations still differ. Users do not experience the no-confirmation scenario in Codex, as it's an internal research configuration.*

17 For ScreenSpot-Pro and ExploitGym, the Fable scores we report come from Mythos, which is Fable with fewer safeguards.

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